ggtree_custom_recipes

`ggtree` Custom Recipes

We will load a phylogenetic tree in Newick format for demonstration.

Prepare session:

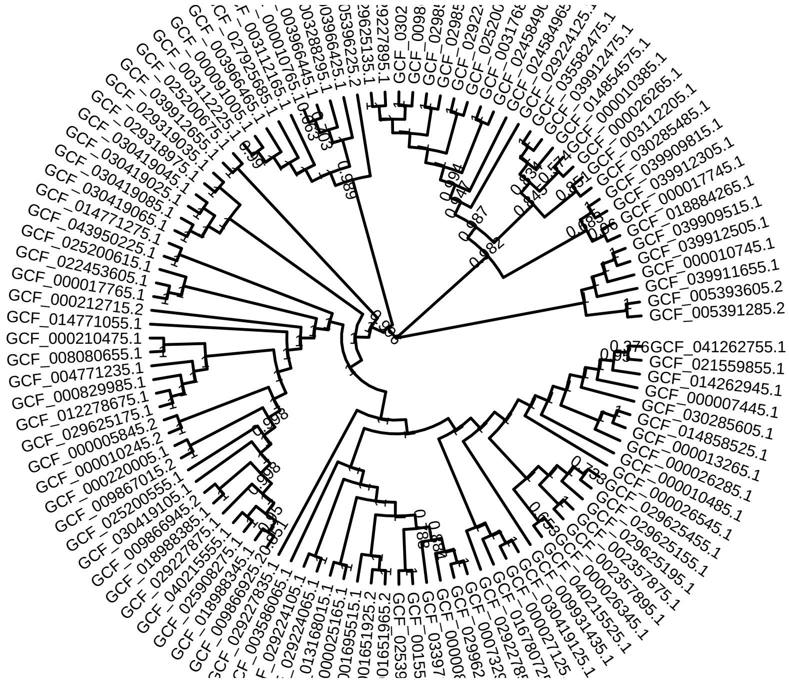
```r
library(ggtree)
library(treeio)
library(ggplot2)
library(dplyr)
```

```r
# Load a tree file
tree <- read.newick("data/example_2_phylo.nwk")
tree$node.label <- round(as.numeric(tree$node.label), digits = 3)
```

Circular Layout Visualization

Circular layouts are excellent for presenting large or complex trees. They allow for efficient space usage and a visually appealing presentation.

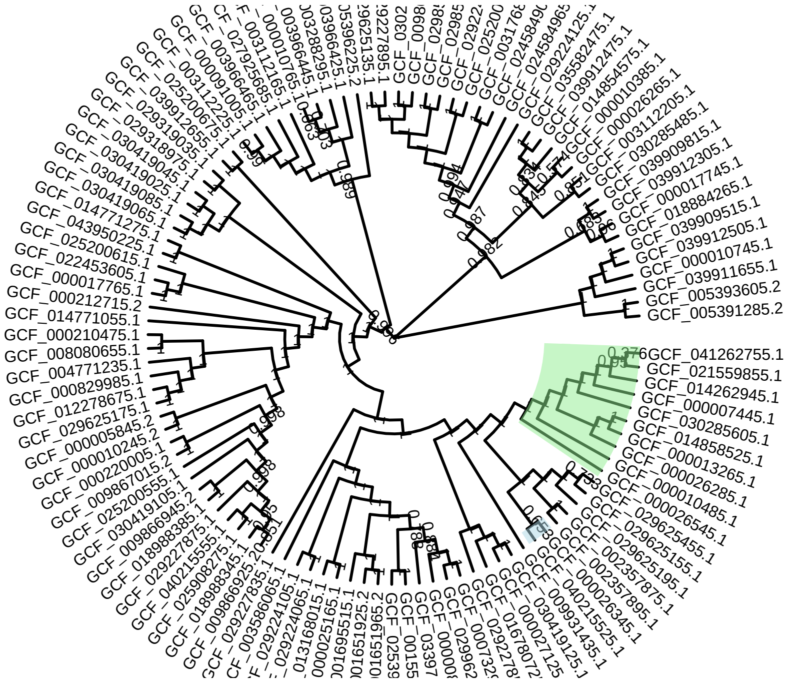
```r
# Circular tree with basic styling
circular_tree <- ggtree(tree, layout = "circular",
                        branch.length = "none") +
  geom_tiplab(size = 2, aes(angle = angle), offset = 0.5) +
  geom_nodelab(color = "black", size = 2, nudge_y = 0.5)

circular_tree
```



Highlighting Clades in Circular Trees

Highlighting specific clades can make the tree more interpretable. Use the `geom_hilight` function:

```
# Highlight specific clades
highlight_tree <- circular_tree +
  geom_hilight(node = 199, fill = "lightblue", alpha = 0.5) +
  geom_hilight(node = 200, fill = "lightgreen", alpha = 0.5)

highlight_tree
```



Adding Heatmaps to Circular Trees

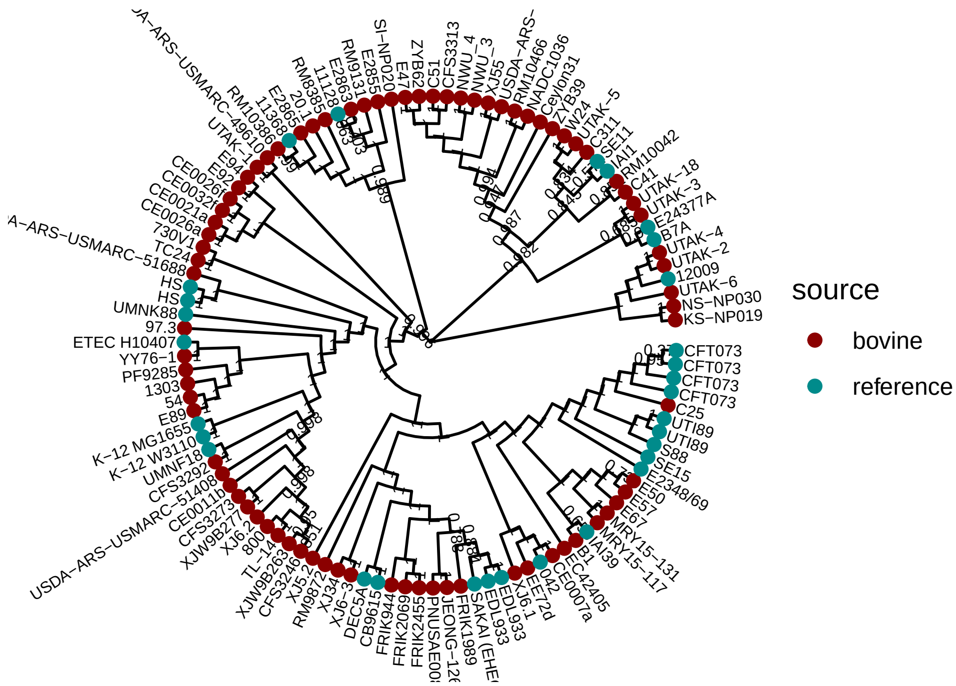
We can add metadata as heatmaps around the tree:

```r
# Cargar metadata
metadata <- readr::read_tsv("data/Experiment_Design_example_2.tsv")
```

```
Rows: 110 Columns: 50
-- Column specification --------------------------------------------------------
Delimiter: "\t"
chr  (37): Assembly.Accession, Assembly.Level, Strain, Attribute, Organism.N...
dbl  (11): ANI.Best.ANI.match.ANI, ANI.Best.ANI.match.Assembly.Coverage, ANI...
lgl   (1): find
dttm  (1): Last.updated

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
colnames(metadata)[1] <- "label"

# The rows have to be in the same order as the tree
```

```
metadata_ordered <- metadata %>%
  filter(label %in% tree$tip.label) %>%
  arrange(match(label, tree$tip.label))

metadata_ordered$Phylogroup <- replace(metadata_ordered$Phylogroup, is.na(metadata_ordered



#Convert tree information to tibble object and add metadata
treetib <- as_tibble(tree)
treetib <- left_join(treetib, metadata_ordered, by = c('label'))

tree_ec <- as.treedata(treetib)




# Merge metadata with tree
p_circular_heatmap <-  ggtree(tree_ec, layout = "circular",
                        branch.length = "none") +
  geom_tiplab(size = 2, aes(label = Strain,angle = angle), offset = 0.5) +
  geom_nodelab(color = "black", size = 2, nudge_y = 0.5)+
  geom_tippoint(aes(color = source), size = 2) +
  scale_color_manual(values = c("darkred", "darkcyan"))

p_circular_heatmap
```

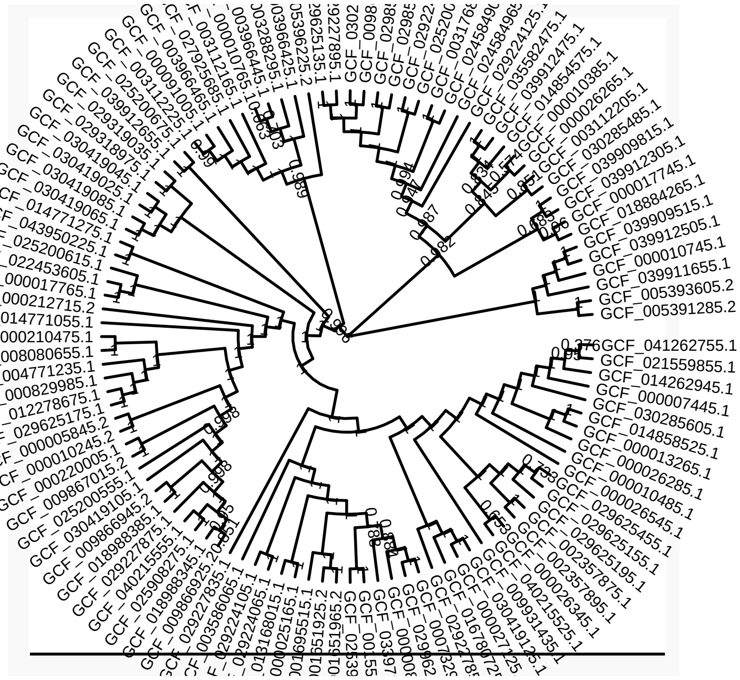


Styling Trees with Advanced Themes

`ggtree` provides options to customize the appearance of trees using themes. Here's how to use them effectively.

Customizing Themes for Trees

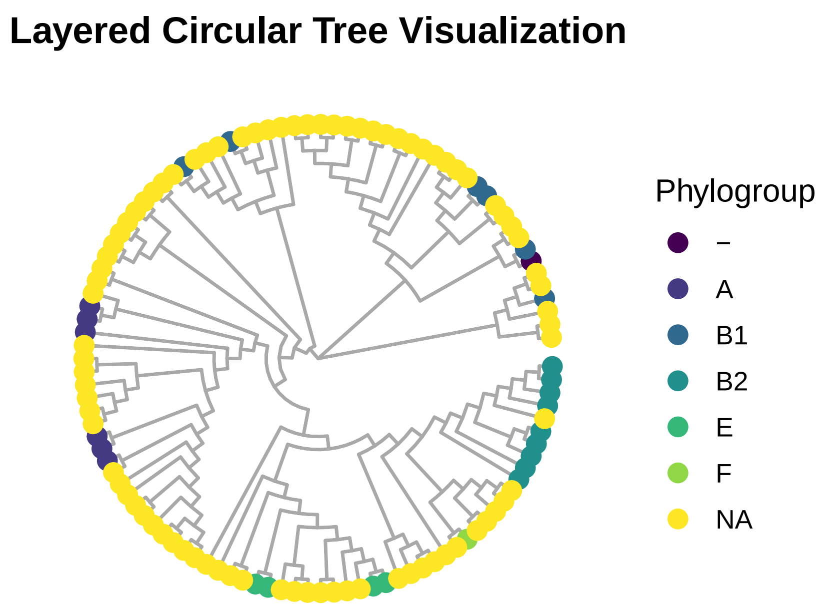
```
# Applying a theme with a clean background
styled_tree <- circular_tree +
  theme_tree2() +
  theme(
    plot.background = element_rect(fill = "#f9f9f9", color = NA),
    panel.border = element_blank(),
    panel.grid = element_blank(),
    axis.text = element_blank(),
    axis.ticks = element_blank()
  )

styled_tree
```



Adding Multiple Aesthetic Layers

Combine multiple layers for richer visuals:

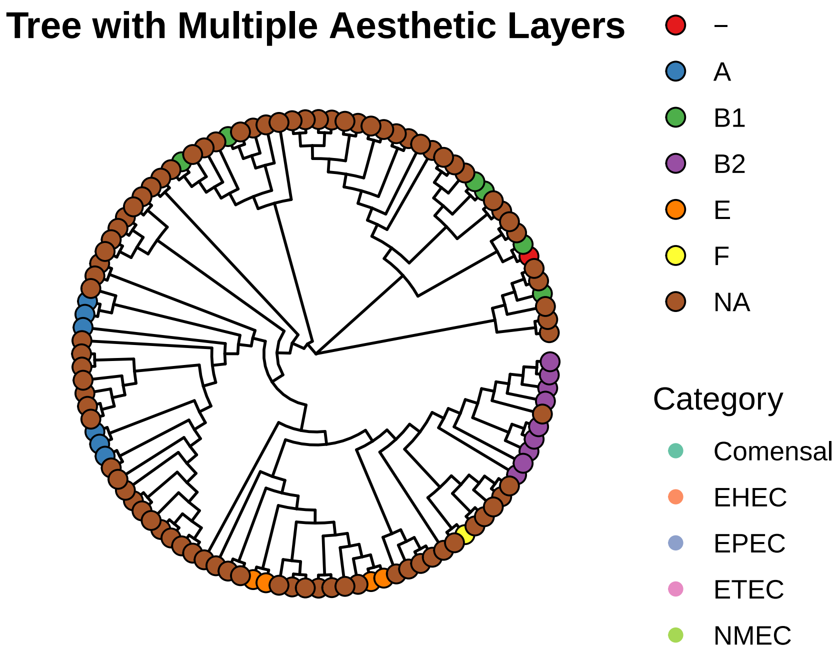
```r
# Adding node points and branches with customized colors
layered_tree <- ggtree(tree_ec, layout = "circular", branch.length = "none") +
  geom_tree(color = "darkgrey", size = 0.6) +
  geom_tippoint(aes(color = Phylogroup), size = 3) +
  scale_color_viridis_d() +
  theme(
    legend.position = "right",
    legend.title = element_text(size = 12),
    legend.text = element_text(size = 10),
    plot.title = element_text(size = 14, hjust = 0.5, face = "bold")
  ) +
  ggtitle("Layered Circular Tree Visualization")

layered_tree
```



Combining Trees with ggnewscale

Use the `ggnewscale` package to add multiple scales of colors:

```
# Adding new scale for different metadata layers
multi_scale_tree <- ggtree(tree_ec, layout = "circular", branch.length = "none") +
  geom_tippoint(aes(color = Patotype), size = 2) +
  scale_color_brewer(palette = "Set2", name = "Category") +
  ggnewscale::new_scale_color() +
  geom_tippoint(aes(fill = Phylogroup), shape = 21, size = 3, color = "black") +
  scale_fill_brewer(palette = "Set1", name = "Fill Category") +
  theme(
    legend.position = "right",
    legend.title = element_text(size = 12),
    legend.text = element_text(size = 10),
    plot.title = element_text(size = 14, hjust = 0.5, face = "bold")
  ) +
  ggtitle("Tree with Multiple Aesthetic Layers")

multi_scale_tree
```



Tree with Multiple Aesthetic Layers
–
A
B1
B2
E
F
NA
Category
Comensal
EHEC
EPEC
ETEC
NMEC